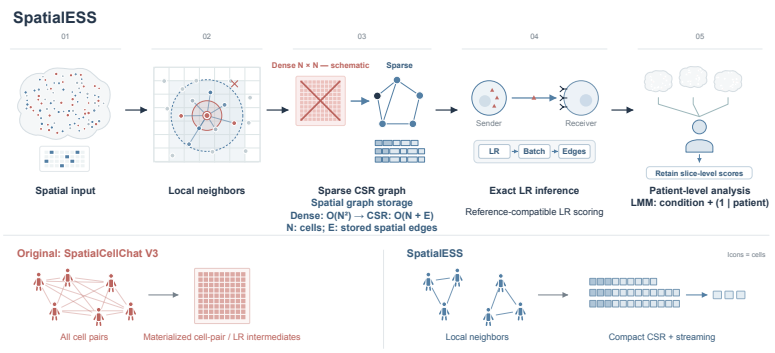
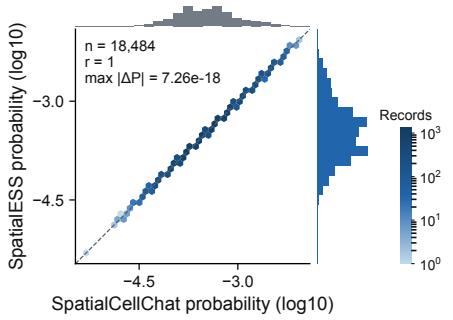


B Probability fidelity · 50k

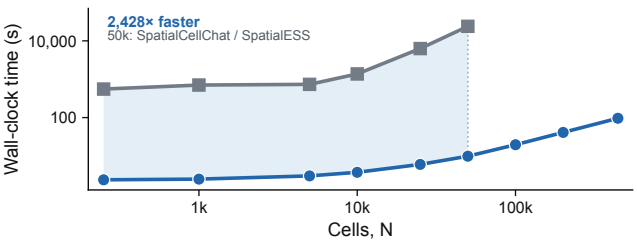


C Fidelity across paired scales

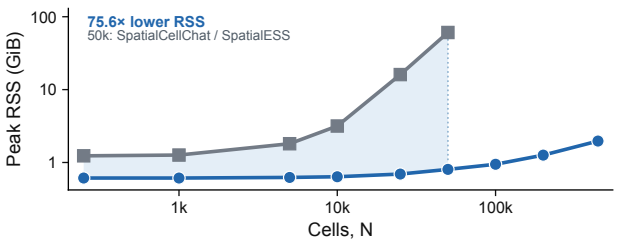
	250	1k	5k	10k	25k	50k
Probability r	1	1	1	1	1	1
Max |Δ probability|	5.96e-19	1.73e-18	2.60e-18	3.69e-18	7.81e-18	7.26e-18
Exact p fraction	1	1	1	1	1	1
Active-record Jaccard	1	1	1	1	1	1
Call disagreements	**0**	**0**	**0**	**0**	**0**	**0**

SpatialESS SpatialCellChat

D Runtime

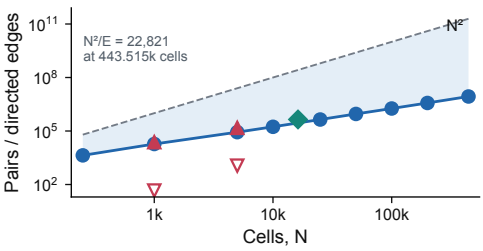


E Peak memory

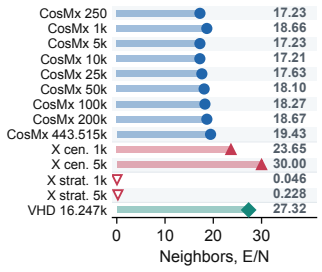


CosMx central Xenium central Xenium stratified Visium HD ovarian

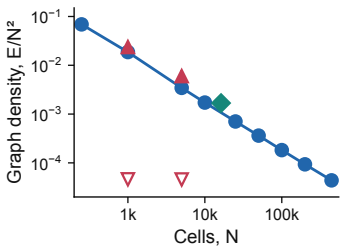
F Pair space and local edges



G Local neighbors per cell



H Graph density



Directed non-self edges; effective radius = 35 µm. Shading in D–F is a comparison guide, not a confidence interval.

I Completion and comparison scope

